Mechatronic Sensors Trainer

Fundamental Labs – Time of Flight

V1.0 – 16th January 2026

QUANSER
INNOVATE · EDUCATE

Quanser Consulting Inc.
119 Spy Court
Markham, Ontario
L3R 5H6, Canada

info@quanser.com
Phone : 19059403575
Fax : 19059403576
printed in Markham, Ontario.

Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Mechatronic Sensors Trainer – Application Guide
Time of Flight Lab

Why Explore Time of Flight?

Measuring distance can be performed using different technologies. As the name suggests, time of flight uses an emitter to send a wave pulse and using the time it takes for the pulse to come back to the emitter, estimate the distance to the object. Exploring time of flight sensors is important for knowing what material surfaces are better suited for distance measurements. In mechatronics systems Time of Flight sensors can be seen in drones, manufacturing cells, gesture identification.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**023_time_of_flight**.

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of the **Time of Flight**. The lab will guide you through understanding single beam and multizone measurements for single photon avalanche diode (SPAD) grid array-based Time of Flight sensor.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.